The nature of econometrics and application of econometric models in economics and business

Econometrics. Lecture 1

Content

1. What is econometrics?
2. Steps in empirical economic analysis
3. The structure of economic data

1. What is econometrics?

Economic model links one or more economic variables to other economic variables

What impact on price growth can be expected from the Central Bank's interest rate hike?

What matters most to decision-makers is not the explanation of the relationship, but the magnitude of the association between the variables.

A key objective of econometrics is to convert qualitative statements into quantitative ones

Qualitative statement

- Higher interest rate reduces inflation

Quantitative statement

- Inflation decreases by 0.3 p.p. a year after a 1 p.p. increase in the interest rate

How to convert a qualitative statement into a quantitative one?

Use **statistical data** …
… and **statistical and mathematical techniques**

ECONOMICS

Theory / questions

What should happen?
Why might it happen?
What mechanisms matter?

STATISTICS

Evidence

What actually happened?
How much variation is there?
How noisy is the evidence?

MATHEMATICS

Estimation

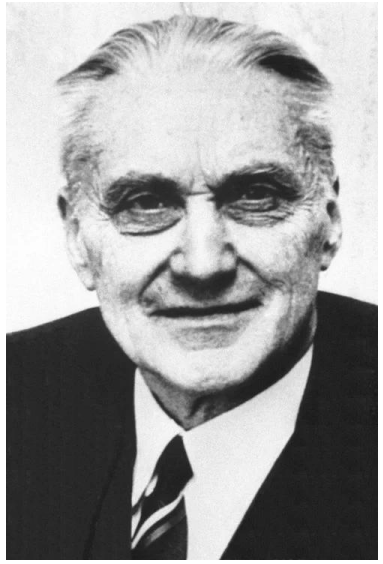
How to quantify?

ECONOMETRICS

Learning

What relationship is in the data?
How large is it?
How certain are we?

What is econometrics? (1)



Ragnar Frisch

Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 1969 (jointly with J. Tinbergen)

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But there are several aspects of the quantitative approach to economics, and no single one of these aspects, taken by itself, should be confounded with econometrics. Thus, econometrics is by no means the same as economic statistics. Nor is it identical with what we call general economic theory, although a considerable portion of this theory has a definitely quantitative character. Nor should econometrics be taken as synonomous with the application of mathematics to economics. Experience has shown that each of these three view-points, that of statistics, economic theory, and mathematics, is a necessary, but not by itself a sufficient, condition for a real understanding of the quantitative relations in modern economic life. It is the *unification* of all three that is powerful. And it is this unification that constitutes econometrics.

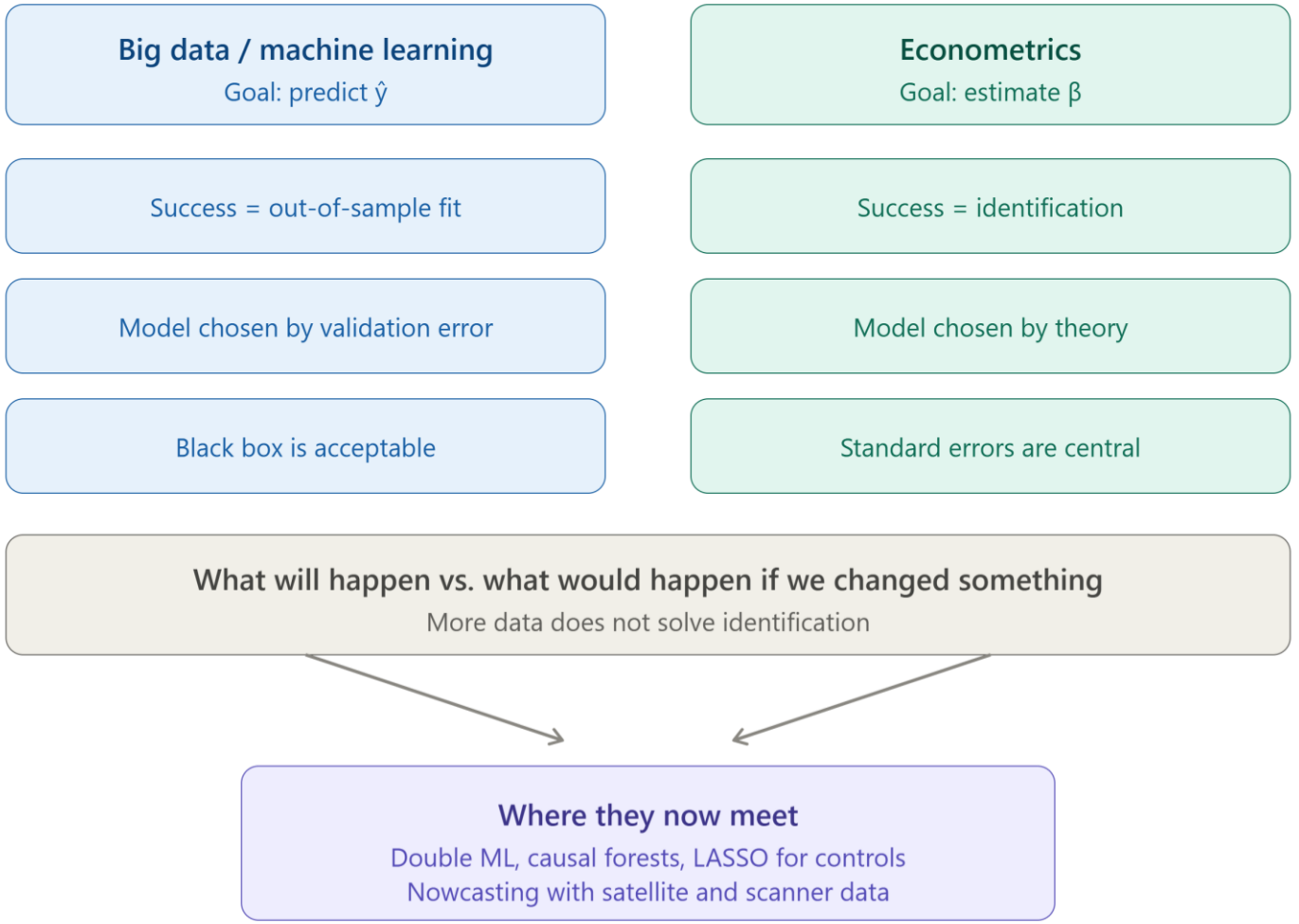
Frisch, R. (1933). Editor's Note. *Econometrica, 1*(1), 1-4.

What is econometrics? (2)

Econometrics combines economic theory, mathematical and economic statistics to

- analyze economic relationships,
- test economic theories,
- evaluate government or business policies.

Machine learning ≠ Econometrics



Economic reasoning is needed to interpret statistical evidence!

Econometrics connects economics and data

Why not mathematical statistics?

Econometrics is different from mathematical statistics due to **nonexperimental** nature of the economic data.

Controlled experiments are the norm in the natural sciences but are often impossible, prohibitively expensive, or ethically unacceptable in economics.

However, even in Econometrics there is a recent trend to estimate models that resemble natural experiments.

Where do we see econometrics?

CENTRAL BANK 

Will raising the policy rate reduce inflation?

GOVERNMENT 

Does a higher minimum wage reduce poverty?

EDUCATION 

Does more education increase earnings?

HEALTH 

Does a new treatment improve outcomes?

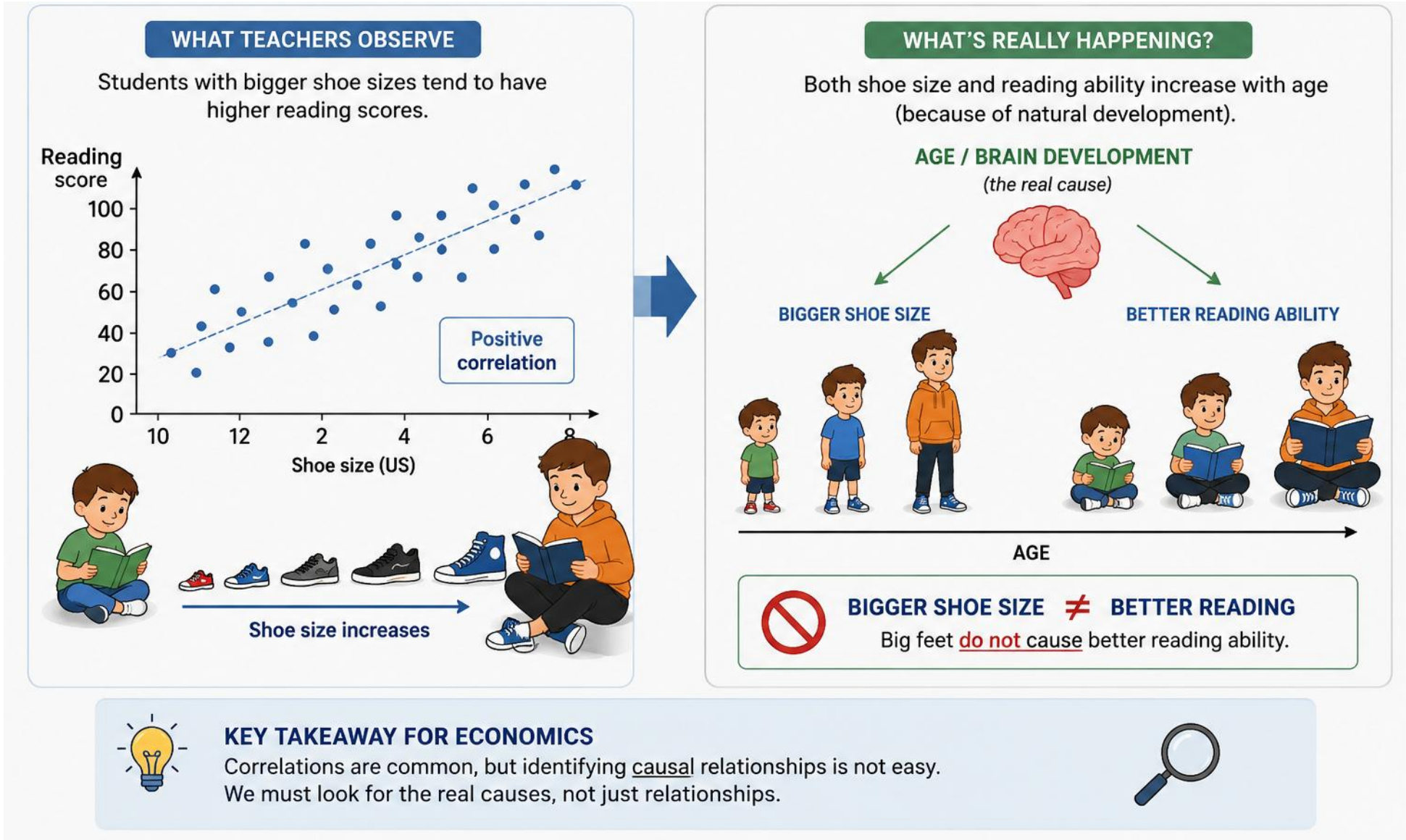
BUSINESS 

Will lowering the price increase sales?

MACROECONOMICS 

What will happen to GDP next year?

Correlation vs. causation – the fundamental issue



A problem economists actually face

Policy rate ↑ → Inflation ↓

But when does the central bank raise rates?

INFLATION
Expected inflation ↑

—

CENTRAL BANK
Interest rate ↑

—

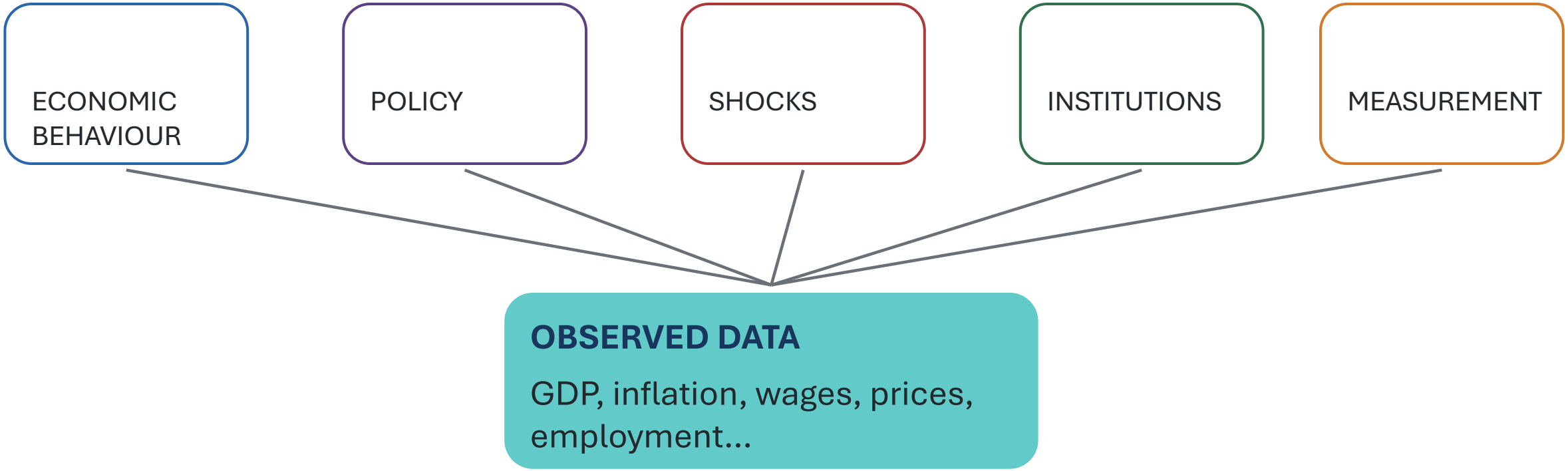
INFLATION
Keeps increasing for some time

—

DATA
Positive correlation?

While the raw data can show a positive correlation, the actual causal effect can be negative (*"without the interest rate hike inflation will increase more"*).

Why is economic data so difficult?



Economic outcomes are generated by many forces at once

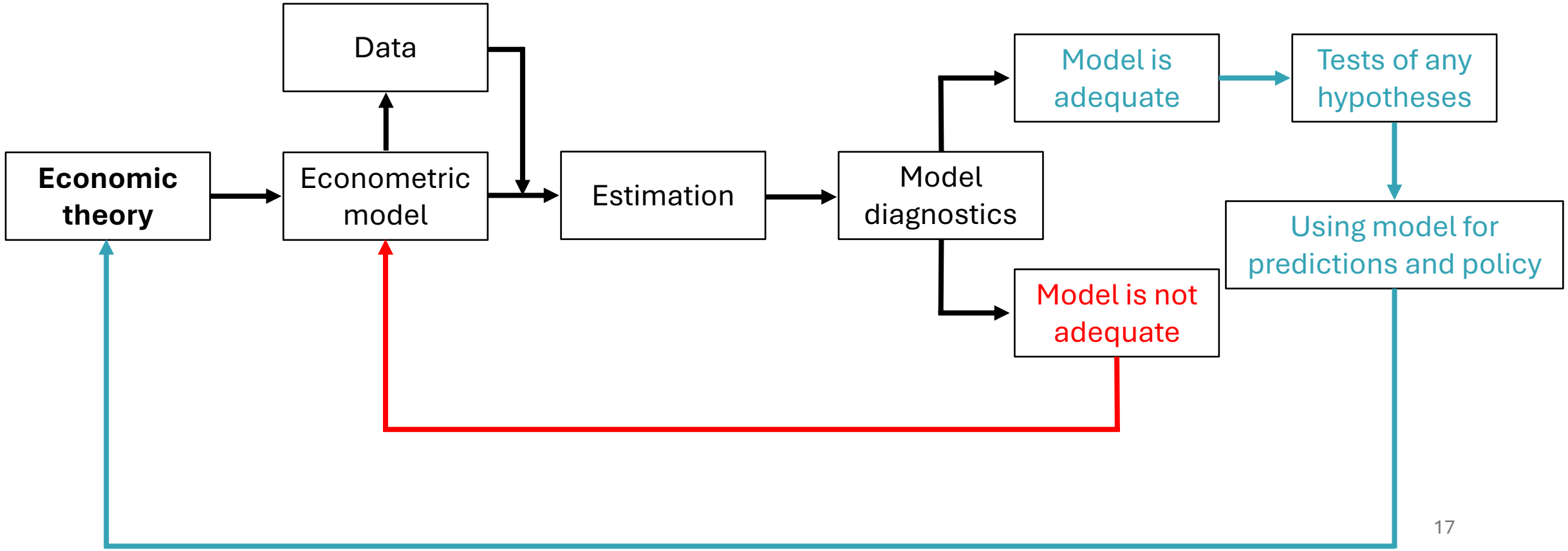
Ceteris paribus

- The economist's goal often is to infer the **causal effect** of one variable (such as education) on another (such as worker productivity).
- An association between two or more variables may suggest, but does not establish, a causal effect (**correlation does not imply causality**).
- **Ceteris paribus** (other factors held constant; all else equal) plays a key role in a causal analysis.
  - E.g. demand analysis focuses on the effect of a price increase of a good on the quantity demanded, holding income, other prices, etc. fixed.
  - If other factors are not held fixed, then we cannot isolate the causal effect of a price change on the quantity demanded.

2. Steps in empirical economic analysis

Steps of the applied econometric analysis

Economists use **empirical analysis** to test their hypotheses and quantify economic relationships



Step 1
Formulation of the question of interest

- Does an increase in government spending have an impact on economic growth?
- What is the impact of policy interest rate change on inflation?
- Do less capitalized banks react differently to the monetary policy shock?
- Does job training have an impact on worker productivity?
- Does an additional year of schooling have an impact on one's wage?
- Is there an impact of legalized abortion on crime?
- Does a price increase have an impact on firm's revenues?

Step 2
Specification of economic model (1)

Sometimes a **formal economic model** is constructed

- Demand theory assumes that individuals allocate their budget to a set of goods such that they maximize their utility for a given set of prices.
- This implies a system of demand equations that relate the quantity consumed of goods to the consumer's budget, prices and individual characteristics that affect taste

$$q_1 = f_1(x, p_1, \dots, p_k, age, educ)$$
$$\vdots$$
$$q_k = f_k(x, p_1, \dots, p_k, age, educ)$$

Step 2
Identification of economic model (2)

Model 1: Determinants of wage $y = f(x_1, x_2, \dots, x_7)$

Question: Suppose that y is wage in Riga and $x_1, \dots, x_7$ are factors affecting it. Give examples of such factors. What is the sign of their effect?

In many cases, empirical analyses make use of economic theory in a less formal way or even rely on **intuition**

- A labor economist is interested in the effect of job training on worker productivity
- Basic economic understanding is sufficient to realize that factors like education, experience and training affect worker productivity
- Moreover, workers are paid according to their productivity
- This leads to a model such as

$$wage = f(educ, exper, training)$$

Step 3
Formulation of an econometric model

- **Form** of the demand equations or the wage equation has to be specified
- One must decide how to deal with **unobserved variables**

A complete **econometric model** might look like

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 training + u$$

where “educ” and “exper” are years of education and experience, and “training” is whether a person had additional training.

- β_0 , β_1 , β_2 and β_3 are called the **parameters** (of the model)
- The term u is the **error term** or **disturbance term** and contains factors such as innate ability, quality of education and all other unobserved factors that influence one’s wage

Step 4
Data collection

Data on relevant variables have to be collected

- Microeconomic data (surveys: individuals' consumption of a set of goods, prices, budgets and characteristics, workers' wages, education, experience, training, business performance indicators, gender data, etc.)
- Macroeconomic data (statistical offices, central banks, IMF, World Bank, OECD, CEIC, etc.)
- Financial data (commercial banks, Bloomberg, Reuters, stock exchanges, etc.)

Step 5
Estimation and diagnostics of the model

Econometric methods can then be used to estimate the econometric model's parameters and to formally test hypotheses of interest.

After the estimation of the model is done, the applied econometrician has to perform **specification tests** to make sure that the model she/he used was the appropriate one, as well as some **diagnostic checking** in order to check the performance and the accuracy of the estimation procedure.

Step 6
Test hypotheses and interpret your model

If the model is adequate, then the next test is to apply hypothesis testing in order to test the validity of the theoretical predictions, and then she/he will be able to use the model for making predictions and policy recommendations.

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 training + u$$

- We may hypothesize that training increases a worker's wage
- The latter hypothesis is equivalent to $\beta_3 > 0$ in the wage equation

Carefully interpret the results. Are they **economically meaningful**?

Step 7
Present results

- Journal article
- Research paper
- Policy report
- Analytical note

3. The structure of economic data

Data for econometrics

A **cross-sectional data** set consists of a sample of individuals, households, firms, cities, states, countries, or a variety of other units, taken at a given point in time.

Time series data track variables over time (stock prices, money supply, consumer price index, gross domestic product).

Panel data consist of a time series for each member in the data set (e.g. we may have GDP from 1991 till 2025 for all Eastern European countries).

Cross-sectional data

Cross-sectional variables are usually denoted by the subscript *i*, with *i* taking values from 1, 2, 3, … , *N*; for *N* number of cross-sections. If *Y* denotes the income data we have collected for *N* number of individuals, this variable, in a cross-sectional framework, will be denoted by:

$$Y_i \quad \text{for } i = 1,2,3, \dots, N$$

An important feature of cross-sectional data is that we can often assume that they have been obtained by **random sampling** from the underlying population (for survey data).

Ordering of the data does not matter for the analysis.

Cross-sectional data: Examples

Table 1. Wages and other individual characteristics

obsno	wage	educ	exper	female	married
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
⋮	⋮	⋮	⋮	⋮	⋮
526	3.50	14	5	1	0

Table 2. GDP per capita, life expectancy and spending on research in 2021

obsno	country	gdp_ppp	life_exp	RandD_share
1	Belarus	27 611.18	72.07	0.46
2	Lithuania	45 873.53	74.04	1.10
⋮	⋮	⋮	⋮	⋮
120	Germany	65 530.77	80.79	3.13

Time-series data

Time series data are denoted with the subscript *t*. So, for example, if *Y* denotes the GDP of a country from 1990 to 2025 then we denote that as:

$$Y_t \quad \text{for } t = 1,2,3, \dots, T$$

where $t = 1$ for 1990 and $t = T = 36$ for 2025.

- Observations can rarely be assumed to be independent across time.
- Data frequency can be daily, weekly, monthly, quarterly, annual, etc.
- Many weekly, monthly, and quarterly economic time series display a strong seasonal pattern, which can be an important factor in a time series analysis.

Time-series data: Example

Table 3. US macroeconomic data

period	rgdp	cpi	fedrate	eurusd	oil_price
2000Q1	13 878.15	100.00	5.68	0.99	26.85
2000Q2	14 130.91	100.70	6.27	0.93	26.52
2000Q3	14 145.31	101.52	6.52	0.91	30.20
⋮	⋮	⋮	⋮	⋮	⋮
2026Q2	24 270.60	194.48	3.63	1.17	104.43

Time series data are mainly associated with macroeconomic applications.

Pooled cross sections

Such data have both **cross-sectional** and **time series** features.

For example:

- Random sample of households in 2015 with variables like income, savings, family size, etc.
- New random sample of households in 2025 with the same variables as in 1985.
- Pooled cross section combines both years

In a pooled cross section, the households observed are **different**.

Pooled cross sections: Example

Table 4. Two years of housing prices

obsno	year	hprice	proptax	sqrft	bdrms	bthrms
1	1993	85 500	42	1 600	3	2.0
2	1993	67 300	36	1 440	3	2.5
⋮	⋮	⋮	⋮	⋮	⋮	⋮
251	1995	65 000	16	1 250	2	1.0
252	1995	182 400	20	2 200	4	2.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
520	1995	57 200	16	1 100	2	1.5

Pooling cross sections from different years is often an effective way of analyzing the effects of a new government policy. The idea is to collect data from the years before and after a key policy change.

Panel data

Panel data are denoted by the use of both *i* and *t* subscripts. This is simply because panel data have **both cross-sectional and time series dimensions:**

$$Y_{it} \quad \text{for } t = 1,2,3,\dots,T \quad \text{and } i = 1,2,3,\dots,N$$

A **panel data** set consists of a time series for each cross-sectional member in the data set.

Key distinction with pooled cross sections is that the **same cross-sectional units** (individuals, firms, countries, etc.) are followed over a given time period

Panel data: Example

Table 5. Data on commercial banks

date	bank	Interbank_rate	lending_rate	npl	liquidity	assets
2013m1	108	35.3	37.1	4.3	52.1	451 458
2013m1	110	35.3	43.2	1.6	76.9	47 135
⋮	⋮	⋮	⋮	⋮	⋮	⋮
2015m1	108	45.2	51.9	12.4	224.1	760 364
2015m1	110	45.2	50.9	4.0	57.1	81 192
⋮	⋮	⋮	⋮	⋮	⋮	⋮
2017m12	108	9.7	10.6	21.8	74.2	757 523
2017m12	110	9.7	12.9	4.5	131.2	109 398

Specific econometric techniques exist to analyze panel data.

Summary.
What do econometricians actually do?

1 DESCRIBE

What relationship do we see?

How strong is it?
How much variation is there?
What does the data look like?

2 IDENTIFY

What can we say about the underlying economic mechanism?

Can we interpret an effect causally?

3 PREDICT

What is likely to happen next?

How uncertain is the forecast?
What scenarios are plausible?

Econometrics is the art and science of learning about economic relationships from data

Summary.
Economics, statistics and econometrics

ECONOMICS	Why might interest rate affect inflation?
STATISTICS	Are interest rate and inflation associated in the data?
ECONOMETRICS	What can we learn about the economic relationship from the data? Does interest rate actually cause lower inflation?

Economics without data is untested. Statistics without theory is pattern-matching. Econometrics is the discipline of making the two speak to each other.

Let's play!
Simulation of the Central Bank Board Meeting

You: Forecasting team at the Central Bank.

Your task: propose and justify to the board a decision on the key interest rate.

Some technical assumptions *(for the knowledge of which econometric models are necessary)*:

- Inflation depends on economic activity, the exchange rate, and unexpected shocks.
- A 1 p. p. increase in the key rate and its maintenance at this level leads to a 1.2 p. p. decrease in inflation after one year (the opposite is true for a rate reduction).
- An increase in economic activity above the sustainable level by 1% leads to a 0.7 p. p. increase in inflation after one year (the opposite is true for a reduction).
- A 1% weakening of the national currency increases inflation by 0.4 p.p. after one year (the opposite is true for an appreciation).
- **You can change the interest rate in increments of 0.25 p. p.**

Simulations are based on macroeconomic gap model, which is currently used for economic analyses and forecasting

Kharitonchik, A. (2024) Optimal monetary policy framework in an emerging market economy under sanctions pressure and restrictions on capital flows. *Central European Economic Journal*,*11(58)*, 329-345. DOI: https://doi.org/10.2478/ceej-2024-0022.

We will discuss what your decision will lead to in the country's economy 😁

Scenario 1

A major trading partner is experiencing an economic boom. Your country’s firms have increased exports and are operating at maximum capacity.

Wages have risen significantly, leading to a strong increase in consumer demand in your country.

As a result, your country's economic activity level is **2% above its optimal level**.

The national currency has **appreciated by 2%** due to significant export revenues.

Scenario 2

The summer was very dry and hot. The grain harvest was the lowest in a decade.

Experts at your central bank predict that due to the poor harvest, inflation will **accelerate by 2 percentage points in a year**.

Your central bank experts also believe that inflation expectations in the country have increased due to rising prices for everyday products, and this factor will **add another 1 percentage point** to inflation over the year.

Scenario 3

Your economy is going through tough times. Economic activity contracted (GDP is currently **2% below its sustainable level**) due to past economic policy mistakes.

Consumer sentiment is at an extremely low level.

Experts at your central bank believe that inflation expectations are currently near historical lows and will have a **disinflationary impact of 1 percentage point** over a one-year horizon.

Scenario 4

Structural reforms in your economy have begun to yield results: labor productivity has increased thanks to an influx of foreign investment, technology, and the liberalization of business environment.

Your experts expect the economy to **grow 2 percentage points faster** than before, **thanks to increased production efficiency**.

The exchange rate of your national currency remains stable, and consumer sentiment is positive.

Scenario 5

Global investors have become more cautious about emerging markets and are withdrawing capital from your country.

The national currency has **weakened by 5% this quarter**, and your experts expect it to **weaken by a further 5% next quarter**.

Foreign direct investment has fallen sharply and several planned projects have been postponed. Economic activity is already **1% below** its sustainable level, and your experts expect it to fall to **2% below** next quarter.

Inflation expectations remain anchored for now.

Scenario 6

Ahead of the elections, the government has sharply raised pensions and public sector wages, financed by borrowing.

Household demand has surged: economic activity is now **1% above its sustainable level**.

Investors have grown concerned about the sustainability of public debt. The risk premium on your country has risen by 5 percentage points, and the national currency has **weakened by 1%**.

Inflation expectations remain anchored, but your experts warn they may not stay that way.